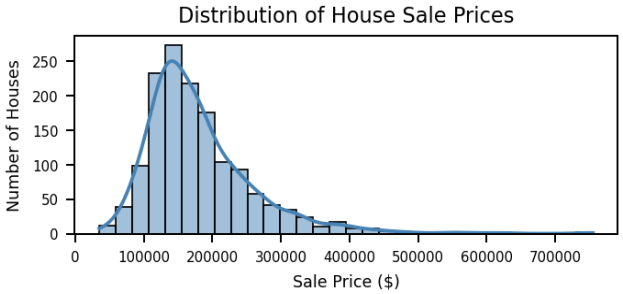
Assignment 2 – Part 3: House Price Data Analysis

Dataset overview: The dataset contains 1,460 houses and 81 variables. The target variable is *SalePrice*. Mean sale price: $180,921; median: $163,000; standard deviation: $79,443.

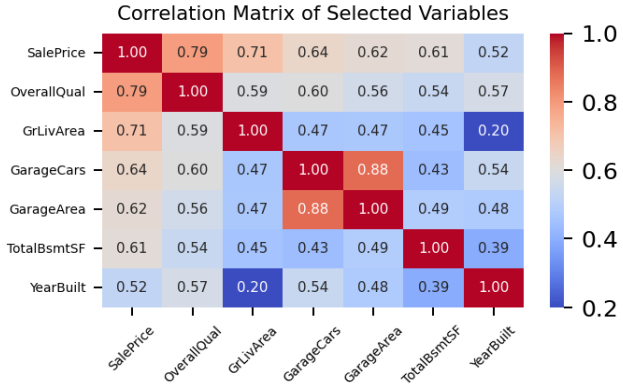
Finding 1: Distribution of SalePrice

Sale prices range from $34,900 to $755,000. The mean ($180,921) is higher than the median ($163,000), and skewness is 1.88. The histogram shows a right-skewed distribution: most homes are concentrated at lower-to-middle prices, while a smaller number of expensive homes form a long right tail.



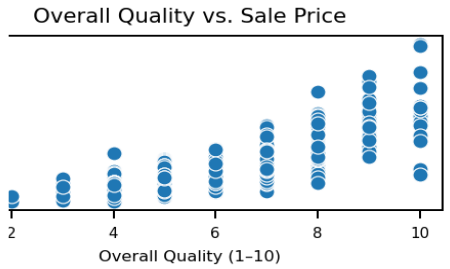
Finding 2: Correlations with SalePrice

Among the numerical variables, OverallQual has the strongest positive correlation with SalePrice (r = 0.79), followed by GrLivArea (r = 0.71). GarageCars (r = 0.64), GarageArea (r = 0.62), and TotalBsmtSF (r = 0.61) are also positively correlated. The heatmap summarizes relationships among the selected variables.



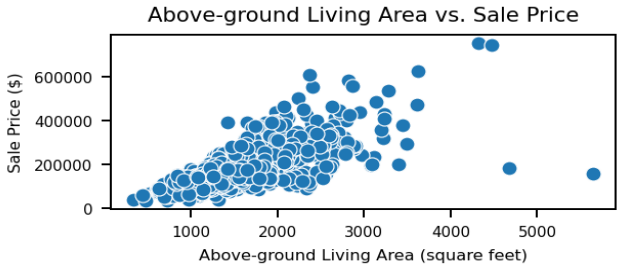
Finding 3: Overall Quality and SalePrice

The scatter plot shows a positive relationship between OverallQual an (r = 0.79). In general, houses with higher overall quality ratings tend to sale prices, although prices still vary within each quality level.



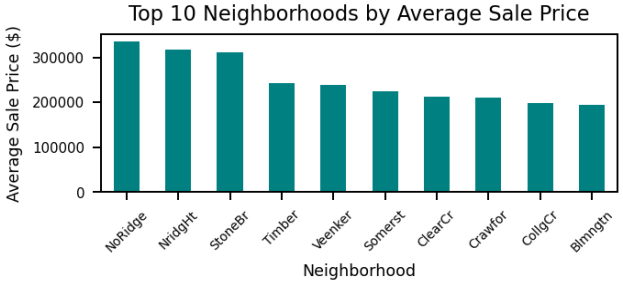
Finding 4: Living Area and SalePrice

GrLivArea has a positive correlation with SalePrice (r = 0.71). Larger above-ground living areas generally tend to be associated with higher sale prices. However, the scatter plot also shows variation and some unusual observations, so area is not the only factor related to price.



Finding 5: Neighborhood and Average SalePrice

Average sale prices differ across neighborhoods. In the 10 neighborhoods with the highest mean prices, NoRidge has the highest average ($335,295). Blmngtn has the lowest average among these top 10 ($194,871). This suggests that location is also associated with house prices.



Overall Conclusion

The EDA suggests that SalePrice is most strongly associated with OverallQual (r = 0.79) and GrLivArea (r = 0.71). Garage-related variables and basement area also show positive correlations. SalePrice is right-skewed, and average prices vary across neighborhoods. These are descriptive associations and do not establish cause and effect.